Reducibility 05

Thumrongsak Kosiyatrakul
tkosiyat@cs.pitt.edu

$REGULAR_{\text{TM}}$ is Undecidable

- **Problem**: Is there an algorithm to determine whether the language of a TM is regular?
- Formally,

$$REGULAR_{\text{TM}} = \{\langle M \rangle \mid M \text{ is a TM and } L(M) \text{ is a regular language}\}$$

- Unfortunately, no algorithm can solve this problem
- To verify this, we need to prove that $REGULAR_{\text{TM}}$ is undecidable
- As usual, we are going to use the proof by contradiction
  - Assume that $REGULAR_{\text{TM}}$ is decidable
  - Thus, there exists a TM R that decides $REGULAR_{\text{TM}}$
    - This gives us the ability to check whether the language of a TM M' is regular or not regular
  - As usual, we want to show that if $REGULAR_{\text{TM}}$ is decidable, A_{TM} is decidable
  - So, we need to construct a TM M' such that whether the language of TM M' is regular or not depending on whether M accepts w

$REGULAR_{\text{TM}}$ is Undecidable

- Once again, we have two choices:
  1. Choice #1:
     - If M accepts w, $L(M')$ is regular
     - If M does not accept w, $L(M')$ is not regular
  2. Choice #2:
     - If M accepts w, $L(M')$ is not regular
     - If M does not accept w, $L(M')$ is regular
- Which one?
- This is an example of the problem where you can pick any choices
  - Let's look at how to construct a TM M' that satisfies each choice

$REGULAR_{\text{TM}}$ is Undecidable

Choice #1

- If M accepts w, $L(M')$ is regular
- If M does not accept w, $L(M')$ is not regular

- Here is an example of a TM M' that satisfies choice #1:

  $M' =$ "On input x:

  1. If x is in the form of 0^n1^n, *accept*.
  2. If x is not in the form of 0^n1^n, run M on input w.
  3. If M accepts w, *accept*. If M rejects w, *reject*."
- From the definition of the above TM M':
  - If M accepts w, $L(M') = \Sigma^*$
  - If M does not accept w:
    - If M rejects w, $L(M') = \{0^n1^n \mid n \geq 0\}$
    - If M loops indefinitely on w, $L(M') = \{0^n1^n \mid n \geq 0\}$

$REGULAR_{\text{TM}}$ is Undecidable

Choice #2

- If M accepts w, $L(M')$ is not regular
- If M does not accept w, $L(M')$ is regular

- Here is an example of a TM M' that satisfies choice #2:

  $M' =$ "On input x:

  1. Run M on input w.
  2. If M accepts w and x is in the form of 0^n1^n, *accept*; otherwise, *reject*.
  3. If M rejects w, *reject*."
- From the definition of the above TM M':
  - If M accepts w, $L(M') = \{0^n1^n \mid n \geq 0\}$
  - If M does not accept w:
    - If M rejects w, $L(M') = \emptyset$
    - If M loops indefinitely on w, $L(M') = \emptyset$

$REGULAR_{\text{TM}}$ is undecidable (Choice #1)

Proof: Assume that $REGULAR_{\text{TM}}$ is decidable. Since $REGULAR_{\text{TM}}$ is decidable, there exists a TM R that decides $REGULAR_{\text{TM}}$. Let construct a TM S as follows:
$S =$ "On input $\langle M, w \rangle$ where M is a TM and w is a string:

1. Construct TM M' as follows:
   $M' =$ "On input x:
   1. If x is in the form of 0^n1^n, *accept*.
   2. If x is not in the form of 0^n1^n, run M on input w.
   3. If M accepts w, *accept*. If M rejects w, *reject*."
2. Run R on input $\langle M' \rangle$.
3. If R accepts $\langle M' \rangle$, *accept*. If R rejects $\langle M' \rangle$, *reject*."

Next, we need to prove that TM S is a decider for A_{TM}.

- Prove that if $\langle M, w \rangle \in A_{\text{TM}}$, S accepts $\langle M, w \rangle$: Assume that $\langle M, w \rangle \in A_{\text{TM}}$. Since $\langle M, w \rangle \in A_{\text{TM}}$, M is a TM that accepts w. Since M accepts w, from the definition of TM M', $L(M')$ is Σ^* which is a regular language. Since $L(M')$ is regular, $\langle M' \rangle \in REGULAR_{\text{TM}}$. Since R is a decider for $REGULAR_{\text{TM}}$, by running R on input $\langle M' \rangle$, R will accept $\langle M' \rangle$. Since R accepts $\langle M' \rangle$, S accepts $\langle M, w \rangle$.
- Prove that if $\langle M, w \rangle \notin A_{\text{TM}}$, S rejects $\langle M, w \rangle$: Assume that $\langle M, w \rangle \notin A_{\text{TM}}$. Since $\langle M, w \rangle \notin A_{\text{TM}}$, M is a TM that does not accept w. Since M does not accept w, from the definition of TM M', $L(M')$ is $\{0^n1^n \mid n \geq 0\}$ which is not a regular language. Since $L(M')$ is not regular, $\langle M' \rangle \notin REGULAR_{\text{TM}}$. Since R is a decider for $REGULAR_{\text{TM}}$, by running R on input $\langle M' \rangle$, R will reject $\langle M' \rangle$. Since R rejects $\langle M' \rangle$, S rejects $\langle M, w \rangle$.

This show that TM S is a decider for A_{TM}. Therefore, A_{TM} is decidable $\rightsquigarrow$ contradiction. Therefore, $REGULAR_{\text{TM}}$ is undecidable.

More Fact for Contradiction

- So far, we have been using A_{TM} as the mean of contradiction
  - Because we know from the fact that A_{TM} is undecidable
- But now, we have the following undecidable languages:
  - $A_{\text{TM}} = \{\langle M, w\rangle \mid M \text{ is a TM that accepts } w\}$
  - $HALT_{\text{TM}} = \{\langle M, w\rangle \mid M \text{ is a TM that halts on input } w\}$
  - $E_{\text{TM}} = \{\langle M\rangle \mid M \text{ is a TM and } L(M) = \emptyset\}$
  - $REGULAR_{\text{TM}} = \{\langle M\rangle \mid M \text{ is a TM and } L(M) \text{ is regular}\}$
- We can now use those undecidable languages
  - For example, to show that a language A is undecidable, we can show that if A is decidable, E_{TM} is decidable
- But we can always use A_{TM}

EQ_{TM} is Undecidable

- **Problem**: Is there an algorithm to determine whether languages of two TMs M_1 and M_2 are identical?
- Formally,

  $EQ_{\text{TM}} = \{\langle M_1, M_2 \rangle \mid M_1 \text{ and } M_2 \text{ are TMs and } L(M_1) = L(M_2)\}$
- Unfortunately, EQ_{TM} is undecidable
- As usual, we are going to assume that EQ_{TM} is decidable
  - In other words, given two TMs M_1 and M_2, we know whether $\langle M_1, M_2 \rangle \in EQ_{\text{TM}}$ or not
- For this language, we generally need to construct two TMs M_1 and M_2 from a TM M and a string w
- For this problem, we are going to look at two proofs:
  - Contradict with the fact that A_{TM} is undecidable
  - Contradict with the fact that E_{TM} is undecidable

EQ_{TM} is Undecidable

Prove that $EQ_{\text{TM}} = \{\langle M_1, M_2 \rangle \mid M_1 \text{ and } M_2 \text{ are TMs and } L(M_1) = L(M_2)\}$ is undecidable.

Proof: Assume that EQ_{TM} is decidable. Since EQ_{TM} is decidable, there exists a TM R that decides EQ_{TM}. Let construct a TM S (to decide A_{TM}) as follows:
$S =$ "On input $\langle M, w \rangle$ where M is a TM and w is a string:

1. Construct TM M_1 as follows:
   $M_1 =$ "On input x:
   1. *accept*.
2. Construct TM M_2 as follows:
   $M_2 =$ "On input x:
   1. Run M on input w.
   2. If M accepts w, *accept*. If M rejects w, *reject*."
3. Run R on input $\langle M_1, M_2 \rangle$.
4. If R accepts $\langle M_1, M_2 \rangle$, *accept*. If R rejects $\langle M_1, M_2 \rangle$, *reject*."

Next, we need to prove that TM S is a decider for A_{TM}.

- Prove that if $\langle M, w \rangle \in A_{\text{TM}}$, S accepts $\langle M, w \rangle$:

  Assume that $\langle M, w \rangle \in A_{\text{TM}}$. Since $\langle M, w \rangle \in A_{\text{TM}}$, M is a TM that accepts w. Since M accepts w, from the definition of TM M_1, $L(M_1) = \Sigma^*$ and from the definition of TM M_2, $L(M_2) = \Sigma^*$. In other words, $L(M_1) = L(M_2)$. Since $L(M_1) = L(M_2)$, $\langle M_1, M_2 \rangle \in EQ_{\text{TM}}$. Since R is a decider for EQ_{TM}, by running R on input $\langle M_1, M_2 \rangle$, R will accept $\langle M_1, M_2 \rangle$. Since R accepts $\langle M_1, M_2 \rangle$, S accepts $\langle M, w \rangle$.

EQ_{TM} is Undecidable

Proof: (continue)

- Prove that if $\langle M, w\rangle \notin A_{\mathrm{TM}}$, S rejects $\langle M, w\rangle$:

  Assume that $\langle M, w\rangle \notin A_{\mathrm{TM}}$. Since $\langle M, w\rangle \notin A_{\mathrm{TM}}$, M is a TM that does not accept w. Since M does not accept w, from the definition of TM M_1, $L(M_1) = \Sigma^*$ and from the definition of TM M_2, $L(M_2) = \emptyset$. In other words, $L(M_1) \neq L(M_2)$. Since $L(M_1) \neq L(M_2)$, $\langle M_1, M_2\rangle \notin EQ_{\mathrm{TM}}$. Since R is a decider for EQ_{TM}, by running R on input $\langle M_1, M_2\rangle$, R will reject $\langle M_1, M_2\rangle$. Since R rejects $\langle M_1, M_2\rangle$, S rejects $\langle M, w\rangle$.

This show that TM S is a decider for A_{TM}. Therefore, A_{TM} is decidable $\rightsquigarrow$ contradiction. Therefore, EQ_{TM} is undecidable.

EQ_{TM} is Undecidable

Prove that $EQ_{\mathrm{TM}} = \{\langle M_1, M_2\rangle \mid M_1 \text{ and } M_2 \text{ are TMs and } L(M_1) = L(M_2)\}$ is undecidable.

Proof: Assume that EQ_{TM} is decidable. Since EQ_{TM} is decidable, there exists a TM R that decides EQ_{TM}. Let construct a TM S (to decide E_{TM}) as follows:
$S =$ "On input $\langle M\rangle$ where M is a TM:

1. Construct TM M_{E} as follows:
   $M_{\mathrm{E}} =$ "On input x:
   1. *reject*.
2. Run R on input $\langle M, M_{\mathrm{E}}\rangle$.
3. If R accepts $\langle M, M_{\mathrm{E}}\rangle$, *accept*. If R rejects $\langle M, M_{\mathrm{E}}\rangle$, *reject*."

Next, we need to prove that TM S is a decider for E_{TM}.

- Prove that if $\langle M\rangle \in E_{\mathrm{TM}}$, S accepts $\langle M\rangle$:

  Assume that $\langle M\rangle \in E_{\mathrm{TM}}$. Since $\langle M\rangle \in E_{\mathrm{TM}}$, M is a TM and $L(M) = \emptyset$. From the definition of TM M_{E}, $L(M_{\mathrm{E}}) = \emptyset$. Since $L(M) = \emptyset$ and $L(M_{\mathrm{E}}) = \emptyset$, $L(M) = L(M_{\mathrm{E}})$. In other words, $\langle M, M_{\mathrm{E}}\rangle \in EQ_{\mathrm{TM}}$. Since R is a decider for EQ_{TM}, by running R on input $\langle M, M_{\mathrm{E}}\rangle$, R will accept $\langle M, M_{\mathrm{E}}\rangle$. Since R accepts $\langle M, M_{\mathrm{E}}\rangle$, S accepts $\langle M\rangle$.

EQ_{TM} is Undecidable

Proof: (continue)

- Prove that if $\langle M \rangle \notin E_{\mathrm{TM}}$, S rejects $\langle M \rangle$:

  Assume that $\langle M \rangle \notin E_{\mathrm{TM}}$. Since $\langle M \rangle \notin E_{\mathrm{TM}}$, M is a TM and $L(M) \neq \emptyset$. From the definition of TM M_{E}, $L(M_{\mathrm{E}}) = \emptyset$. Since $L(M) \neq \emptyset$ and $L(M_{\mathrm{E}}) = \emptyset$, $L(M) \neq L(M_{\mathrm{E}})$. In other words, $\langle M, M_{\mathrm{E}} \rangle \notin EQ_{\mathrm{TM}}$. Since R is a decider for EQ_{TM}, by running R on input $\langle M, M_{\mathrm{E}} \rangle$, R will reject $\langle M, M_{\mathrm{E}} \rangle$. Since R rejects $\langle M, M_{\mathrm{E}} \rangle$, S rejects $\langle M \rangle$.

This show that TM S is a decider for E_{TM}. Therefore, E_{TM} is decidable $\rightsquigarrow$ contradiction. Therefore, EQ_{TM} is undecidable.

ONE_{TM} is Undecidable

- **Problem**: Given a TM M, is there an algorithm to check whether M only accept one string ($|L(M)| = 1$)
- Unfortunately, this is another unsolvable problem
- Formally,

$$ONE_{\text{TM}} = \{\langle M \rangle \mid M \text{ is a TM and } |L(M)| = 1\}$$

- To show that the above problem is unsolvable, we just need to show that ONE_{TM} is undecidable
- Use the same method:
  - Show that if ONE_{TM} is decidable, A_{TM} is decidable
  - We need to construct a TM M' from a TM M and a string w

ONE_{TM} is Undecidable

- Two choices as usual:
  1. Choice #1:
     - If $\langle M, w \rangle \in A_{\text{TM}}$, $\langle M' \rangle \in ONE_{\text{TM}}$
       $\rightsquigarrow$ If M accepts w, $|L(M')| = 1$
     - If $\langle M, w \rangle \notin A_{\text{TM}}$, $\langle M' \rangle \notin ONE_{\text{TM}}$
       $\rightsquigarrow$ If M does not accept w, $|L(M')| \neq 1$
  2. Choice #2:
     - If $\langle M, w \rangle \in A_{\text{TM}}$, $\langle M' \rangle \notin ONE_{\text{TM}}$
       $\rightsquigarrow$ If M accepts w, $|L(M')| \neq 1$
     - If $\langle M, w \rangle \notin A_{\text{TM}}$, $\langle M' \rangle \in ONE_{\text{TM}}$
       $\rightsquigarrow$ If M does not accept w, $|L(M')| = 1$
- This is another example where you can pick either choices
- We are going to use the choice #1

ONE_{TM} is Undecidable

Prove that $ONE_{\mathrm{TM}} = \{\langle M \rangle \mid M \text{ is a TM and } |L(M)| = 1\}$ is undecidable.

Proof: Assume that ONE_{TM} is decidable. Since ONE_{TM} is decidable, there exists a TM R that decides ONE_{TM}. Let construct a TM S (to decide A_{TM}) as follows:
$S =$ "On input $\langle M, w \rangle$ where M is a TM and w is a string:

1. Construct TM M' as follows:
   $M' =$ "On input x:
   1. If $x = \varepsilon$, run M on input w. If M accepts w, *accept*; otherwise, *reject*.
   2. If $x \neq \varepsilon$, *reject*.
2. Run R on input $\langle M' \rangle$.
3. If R accepts $\langle M' \rangle$, *accept*. If R rejects $\langle M' \rangle$, *reject*."

Next, we need to prove that TM S is a decider for A_{TM}.

- Prove that if $\langle M, w \rangle \in A_{\mathrm{TM}}$, S accepts $\langle M, w \rangle$:

  Assume that $\langle M, w \rangle \in A_{\mathrm{TM}}$. Since $\langle M, w \rangle \in A_{\mathrm{TM}}$, M is a TM and that accepts w. Since M accepts w, from the definition of TM M', $L(M') = \{\varepsilon\}$. Since $L(M') = \{\varepsilon\}$, $|L(M')| = 1$. In other words, $\langle M' \rangle \in ONE_{\mathrm{TM}}$. Since R is a decider for ONE_{TM}, by running R on input $\langle M' \rangle$, R will accept $\langle M' \rangle$. Since R accepts $\langle M' \rangle$, S accepts $\langle M, w \rangle$.

ONE_{TM} is Undecidable

Proof: (continue)

- Prove that if $\langle M, w\rangle \notin A_{\mathrm{TM}}$, S rejects $\langle M, w\rangle$:

  Assume that $\langle M, w\rangle \notin A_{\mathrm{TM}}$. Since $\langle M, w\rangle \notin A_{\mathrm{TM}}$, M is a TM and that does not accept w. Since M does not accept w, from the definition of TM M', $L(M') = \emptyset$. Since $L(M') = \emptyset$, $|L(M')| \neq 1$. In other words, $\langle M'\rangle \notin ONE_{\mathrm{TM}}$. Since R is a decider for ONE_{TM}, by running R on input $\langle M'\rangle$, R will reject $\langle M'\rangle$. Since R rejects $\langle M'\rangle$, S rejects $\langle M, w\rangle$.

This show that TM S is a decider for A_{TM}. Therefore, A_{TM} is decidable $\rightsquigarrow$ contradiction. Therefore, ONE_{TM} is undecidable.

Conclusion

- So far, to prove that a language A is undecidable, we use proof by contradiction
  - Assume that A is decidable
  - Since A is decidable, there exists a TM R that decides A
  - Then we show that if A is decidable, A_{TM} is decidable
    - Another known undecidable language can be used instead of A_{TM}
  - So, we need to construct a TM S that decides A_{TM} and prove that S is a decider for A_{TM}
    - Given a TM M and a string w, TM S need to construct a string s such that whether $s \in A$ depends on whether M accepts w
    - Then use TM R (a decider for A) as a helper machine to decide A_{TM}
- Try to prove that ONE_{TM} is undecidable using the choice #2.